Lec-0 to Lec 24

Recurrence Relation → S.M (Substitution)
→ R.T.M (Recursive)

⇓

Master thm

⇓

10 sec

10 sec

Thm

2

1

10 sec

Lec_25

Master's Theorem

Solving Recurrence Relation

⇓

1 to 2 mark

$T(n) = T(n-1) + n$, $T(n) = T(n/4) + 2n$

The Master Theorem is a tool used to solve recurrence relations that arise in the analysis of divide-and-conquer algorithms. The Master Theorem provides a systematic way of solving recurrence relations of the form:

T(n) = aT(n/b) + f(n)

where a>=1 and b>1 and f(n) is positive function

any positive func

How to Solve ?

$T(n) = 2T(n/4) + n^3$

i) $a \geq 1$
ii) $b > 1$
iii) $f(n) =$ +ve fu

$f(n) = n^3$
$a = 2$ $b = 4$

$n, n^2, n^3, e^n, n^{\log n}, \log n, \log\log n$

Case1:

T(n)=8T(n/2)+n^2

$T(n) = aT(n/b) + f(n)$

case1:
case2:
case3:
case4:

$a \geq 1$, $b > 1$, $f(n) = +ve$

Solve:

$f(n) = n^2$, $a = 8$, $b = 2$

$f(n)$ | $n^{\log_b a}$

n^2 | $n^{\log_2 8}$

n^2 | $n^{\log_2 2^3}$

n^2 | $n^{3 \log_2 2}$

n^2 | $n^{3 \times 1} = n^3$

$f(n)$ | $n^{\log_b a}$

$\Downarrow$ | $\Downarrow$

n^2 | n^3

$\boxed{T(c) = \theta(n^3)}$

$TC = O(n^3)$

$TC = \Omega(n^3)$

$\boxed{TC = \theta(n^3)}$

Case2:

T(n)=2T(n/2)+n^2

$TC = \theta(n^2)$

$f(n) = n^2, \quad a = 2, \quad b = 2$

$$\begin{array}{c|c} f(n) & n^{\log_b a} \\ n^2 & n^{\log_2 2} \\ n^2 & n \end{array}$$

$\boxed{TC = \theta(n^2)}$

Case 1: $n^2 \mid n^3$
Case 2: $n^2 \mid n$ $\Rightarrow$ Bigger is Answer

Case3:

- **T(n)=2T(n/2)+n**

$f(n) = n$, $a = 2$, $b = 2$

$f(n)$	$n^{\log_b a}$
n	$n^{\log_2 2}$
n	n

$TC = \theta(n \log n)$

multiply By $\log n$ in case of equal

Case4:

- **T(n)=2T(n/2)+nlogn**

$f(n) = n\log n, \quad a=2, \quad b=2$

$TC = n(\log n)^2$

$f(n)$	$n^{\log_b a}$
	$n^{\log_2 2}$
$n\log n$	n

If Right side is smaller By $(\log n)^k$ than multiply By $(\log n)^{k+1}$ that is Answer

$TC = n \times (\log n)^{k+1}$

$TC = n(\log n)^2$

$TC = n(\log n)^2$

Questions:

- Next video

Case 1:
Case 2:
Case 3:
Case 4:

$n^2 | n^3 \Rightarrow \theta(n^3)$

$n^2 | n \Rightarrow \theta(n^2)$

$n | n \Rightarrow \theta(n \log n)$

$n \log n | n \Rightarrow \theta(n (\log n)^2)$

Mark

① $f(n) = n(\log n)^6$, $n^{\log_b a} = n$

$TC = n$

$TC = n(\log n)$

case 4

$f(n)$ | $n^{\log_b a}$

$n(\log n)^6$ | n

Right side is smaller by $(\log n)^k \Rightarrow k = 6$

$n \log n^{k+1} \rightarrow n(\log n^7)$

Thanks